

1. Matched Pair

Shared init $\theta(0)$
Shared samples (x_i, y_i)
Shared weak input x_w
Shared noise & flow

Both-Feature (B)

Strong + Weak cues
 $\rightarrow M_w^B(\tau)$

Weak-Only Control (W)

Strong cue ablated
 $\rightarrow M_w^W(\tau)$

2. Paired Diagnostics

$\Delta(\tau) = M_w^B(\tau) - M_w^W(\tau)$
(gap in weak response)

$d(\tau) = \dot{\Delta}(\tau)$ (relative drift)
 $d > 0$: transfer; $d < 0$: rate supp.

$\Delta(\tau) = P(\tau) - N(\tau)$
(pathwise drift balance)

3. Causal Decision

Gate 1: $\exists t_s : \Delta(t_s) < 0$
(Outcome Suppr.)

Gate 2: $\tau_W^* \in (0, H]$
(Weak Learnable)

Both Met $\rightarrow$ Causal Starvation

Gate Fails $\rightarrow$ Indeterminate